Projection – Lecture Notes

■ 1. WHAT IS PROJECTION?
- Projection transforms points from n-dimensions to m-dimensions ($m < n$).
- In graphics: 3D to 2D conversion (projecting onto a view/screen plane).

Terminology:
• P = Original point
• P' = Projected point
• Projectors = rays connecting P and P'
• Projection Plane (PP) = where projection appears
• COP = Center of Projection (for perspective)
• DOP = Direction of Projection (for parallel)

■ 2. TYPES OF PROJECTION

➤ Parallel Projection:
• Rays are parallel; no vanishing point
• DOP defines direction
• Subtypes: Orthographic, Cavalier, Cabinet

➤ Perspective Projection:
• Rays converge at COP
• Foreshortening occurs (far objects appear smaller)
• Vanishing points:
- 1-point → 1 vanishing axis
- 2-point → 2 vanishing axes
- 3-point → all three axes vanish

■ 3. PARALLEL PROJECTION MATRIX (XY Plane)
From:
$x' = x + \lambda \cos(\beta)$
$y' = y + \lambda \sin(\beta)$
$z' = 0$

Matrix Form (Oblique projection):

$$\begin{bmatrix} x' \\ y' \\ z' \\ w' \end{bmatrix} = \begin{bmatrix} 1 & 0 & \lambda\cos(\beta) & 0 \\ 0 & 1 & \lambda\sin(\beta) & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

➤ Orthographic: $\lambda = 0$, angle $\alpha = 90°$

■ 4. PERSPECTIVE PROJECTION MATRIX

• COP at origin:
$x' = x / (z/d)$, $y' = y / (z/d)$
$\rightarrow w = z/d$